Code No: 113AU **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech II Year I Semester Examinations, August/September - 2022
ELECTRONIC DEVICES AND CIRCUITS
(Common to EEE, ECE, CSE, IT)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
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1.a) Write the diode current equation? Discuss Static and Dynamic resistances.
b) Explain the V-I characteristics of P-N junction diode in detail. [8+7]

2.a) Discuss the principle of operation of Tunnel diode with the help of energy band diagram.
b) What are the applications of varactor diodes? Explain. [8+7]

3.a) What are the important characteristics of a rectifier circuits?
b) Derive the equation for the efficiency of a half wave rectifier circuit. [6+9]

4.a) A bridge rectifier uses four identical diodes having forward resistance of 5Ω each. Transformer secondary resistance is 5Ω and the secondary voltage of 30V (RMS). Determine the DC output voltage for IDC = 200mA and the value of the ripple voltage.
b) Explain Zener as a voltage regulator. [8+7]

5.a) How does a transistor work as an amplifier? Explain.
b) Draw the hybrid model of BJT and explain the significance of each component. [8+7]

6.a) Explain the construction of a BJT and define all current components.
b) Discuss how to determine the h-parameters from transfer characteristics of a BJT. [8+7]

7.a) What is the need for biasing? Discuss the factors that affecting the operating point.
b) What are the different diode compensation methods for stability? Explain. [8+7]

8.a) Explain the working of n-channel enhancement mode MOSFET.
b) Discuss the volt-ampere characteristics of JFET. [8+7]

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